

2025.12.23

MiniMax M2.1: Significantly Enhanced Multi-Language Programming, Built for Real-World Complex Tasks

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MiniMax has been continuously transforming itself in a more AI-native way. The core driving forces of this process are models, Agent scaffolding, and organization. Throughout the exploration process, we have gained increasingly deeper understanding of these three aspects. Today we are releasing updates to the model component, namely MiniMax M2.1, hoping to help more enterprises and individuals find more AI-native ways of working (and living) sooner.

In M2, we primarily addressed issues of model cost and model accessibility. In M2.1, we are committed to improving performance in real-world complex tasks: focusing particularly on usability across more programming languages and office scenarios, and achieving the best level in this domain.

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Key Highlights of MiniMax M2.1:

Exceptional Multi-Programming Language Capabilities

Many models in the past primarily focused on Python optimization, but real-world systems are often the result of multi-language collaboration. In M2.1, we have systematically enhanced capabilities in Rust, Java, Golang, C++, Kotlin, Objective-C, TypeScript, JavaScript, and other languages. The overall performance on multi-language tasks has reached industry-leading levels, covering the complete chain from low-level system development to application layer development.

WebDev and AppDev: A Comprehensive Leap in Capability and Aesthetics

Addressing the widely recognized weakness in mobile development across the industry, M2.1 significantly strengthens native Android and iOS development capabilities.

Meanwhile, we have systematically enhanced the model's design comprehension and aesthetic expression in Web and App scenarios, enabling excellent construction of complex interactions, 3D scientific scene simulations, and high-quality visualization, making vibe coding a sustainable and deliverable production practice.

Enhanced Composite Instruction Constraints, Enabling Office Scenarios

As one of the first open-source model series to systematically introduce Interleaved Thinking, M2.1's systematic problem-solving capabilities have been further upgraded. The model not only focuses on code execution correctness but also emphasizes integrated execution of "composite instruction constraints," providing higher usability in real office scenarios.

More Concise and Efficient Responses

Compared to M2, MiniMax-M2.1 delivers more concise model responses and thought chains. In practical programming and interaction experiences, response speed has significantly improved and token consumption has notably decreased, resulting in smoother and more efficient performance in AI Coding and Agent-driven continuous workflows.

Outstanding Agent/Tool Scaffolding Generalization Capabilities

M2.1 demonstrates excellent performance across various programming tools and Agent frameworks. It exhibits consistent and stable results in tools such as Claude Code, Droid (Factory AI), Cline, Kilo Code, Roo Code, and BlackBox, while providing reliable support for Context Management mechanisms including Skill.md, Claude.md/agent.md/cursorrule, and Slash Commands.

High-Quality Dialogue and Writing

M2.1 is no longer just "stronger in coding capabilities." In everyday conversation, technical documentation, and writing scenarios, it also provides more detailed and structured responses.

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First Impressions

"MiniMax M2.1 is a great example of open-source models like M2.1 that bring frontier performance (and in some cases exceed the frontier) for a wide variety of use cases. Developers deserve choice, and M2.1 provides that much needed choice!"



Eno Reyes

Co-Founder, CTO of Factory AI

"MiniMax M2.1 performed exceptionally well across our internal benchmarks, showing strong results in complex instruction following, reranking, and handling e-commerce tasks. Beyond its general versatility, it has proven to be an excellent model for coding. We are impressed by the team's commitment to a close collaboration with the MiniMax team as we continue to support their latest innovations on the Fireworks platform."



Benny Chen

Co-founder of Fireworks

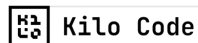
"Minimax M2 series has demonstrated powerful code generation capability, and has quickly become one of the most popular model on Cline platform. We already see another huge advancement in capability for M2.1 and very excited to continue partner with minimax team to explore more use cases."



Saoud Rizwan

Founder, CEO of Cline

"We could not be more excited about M2.1! Our users have come to rely on MiniMax for frontier-grade coding assistance at a fraction of the cost, and early adopters are using it for everything from architecture and orchestration to code reviews and deployment. The speed and efficiency are off the charts."



Scott Breitenother

Co-Founder, CEO of Kilo

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Matt Rubens

Co-Founder, CEO of RooCode

**Robert Rizk**

Co-Founder, CEO of BlackBox AI

"Integrating the MiniMax M2 series into our platform has been a significant win for our users, and M2.1 represents a clear step forward in what a coding-agent can do. We've found that M2.1 handles the nuances of complex, multi-step programming tasks with a level of consistency that is rare in other models. The high-quality reasoning and context awareness at scale, MiniMax has become a core component of how we help developers solve their most challenging problems. We look forward to seeing how our community continues to leverage these updated capabilities."

Benchmarks

MiniMax-M2.1 delivers a significant leap over M2 on core software engineering leaderboards. It shines particularly bright in multilingual scenarios, where it outperforms Claude Sonnet 4.5 and closely approaches Claude Opus 4.5.

Benchmark	MiniMax-M2.1	MiniMax-M2	Claude Sonnet 4.5	Kimi K2 Thinking	DeepSeek V3.2	GLM 4.6	Claude Opus 4.5	Gemini 3 Pro	GPT-5.2 (thinking)
SWE-bench Verified	74.0	69.4	77.2	71.3	73.1	68.0	80.9	78.0	80.0
Multi-SWE-bench	49.4	36.2	44.3	41.9	37.4	30.0	50.0	42.7	x
SWE-bench Multilingual	72.5	56.5	68 ± 0.5	61.1	70.2	53.8	77.5 ± 1.5	65.0	72.0
Terminal-bench 2.0	47.9	30.0	50.0	35.2	46.4	24.5	57.8	54.2	54.0

We also evaluated MiniMax-M2.1 on SWE-bench Verified across a variety of coding agent frameworks. The results highlight the model's exceptional framework generalization and robust stability.

Furthermore, across specific benchmarks—including test case generation, code performance optimization, code review, and instruction following—MiniMax-M2.1 demonstrates comprehensive improvements over M2. In these specialized domains, it consistently matches or exceeds the performance of Claude Sonnet 4.5.

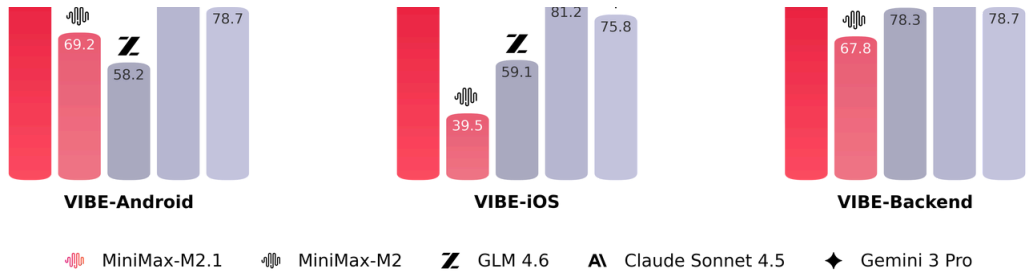
Benchmark	MiniMax-M2.1	MiniMax-M2	Claude Sonnet 4.5	Kimi K2 Thinking	DeepSeek V3.2	GLM 4.6	Claude Opus 4.5	Gemini 3 Pro	GPT-5.2 (thinking)
SWE-bench Verified (Droid)	71.3	68.1	72.3	64.0	67.0	64.9	75.2	x	x
SWE-bench Verified (mini-swe-agent)	67.0	61.0	70.6	63.4	60.0	55.4	74.4	71.8	74.2
SWT-bench	69.3	32.8	69.5	38.2	62.0	45.9	80.2	79.7	80.7
SWE-Perf	3.1	1.4	3.0	1.0	0.9	0.9	4.7	6.5	3.6
SWE-Review	8.9	3.4	10.5	5.3	6.4	5.6	16.2	x	x
OctoCodingbench	26.1	13.3	22.8	16.8	26.0	19.2	36.2	22.9	x

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To evaluate the model's full-stack capability to architect complete, functional applications "from zero to one," we established a novel benchmark: VIBE (Visual & Interactive Benchmark for Execution). This suite encompasses five core subsets: Web, Simulation, Android, iOS, and Backend. Distinguishing itself from traditional benchmarks, VIBE leverages an innovative Agent-as-a-Verifier (AaaV) paradigm to automatically assess the interactive logic and visual aesthetics of generated applications within a real runtime environment. MiniMax-M2.1 delivers outstanding performance on the VIBE aggregate benchmark, achieving an average score of 88.6—demonstrating robust full-stack development capabilities. It excels particularly in the VIBE-Web (91.5) and VIBE-Android (89.7) subsets.

Benchmark	MiniMax-M2.1	MiniMax-M2	Claude Sonnet 4.5	GLM 4.6	Claude Opus 4.5	Gemini 3 Pro
VIBE (Average)	88.6	67.5	85.2	72.9	90.7	82.4
VIBE-Web	91.5	80.4	87.3	86.7	89.1	89.5
VIBE-Simulation	87.1	77.0	79.1	82.4	84.0	89.2
VIBE-Android	89.7	69.2	87.5	58.2	92.2	78.7
VIBE-iOS	88.0	39.5	81.2	59.1	90.0	75.8
VIBE-Backend	86.7	67.8	90.8	78.3	98.0	78.7

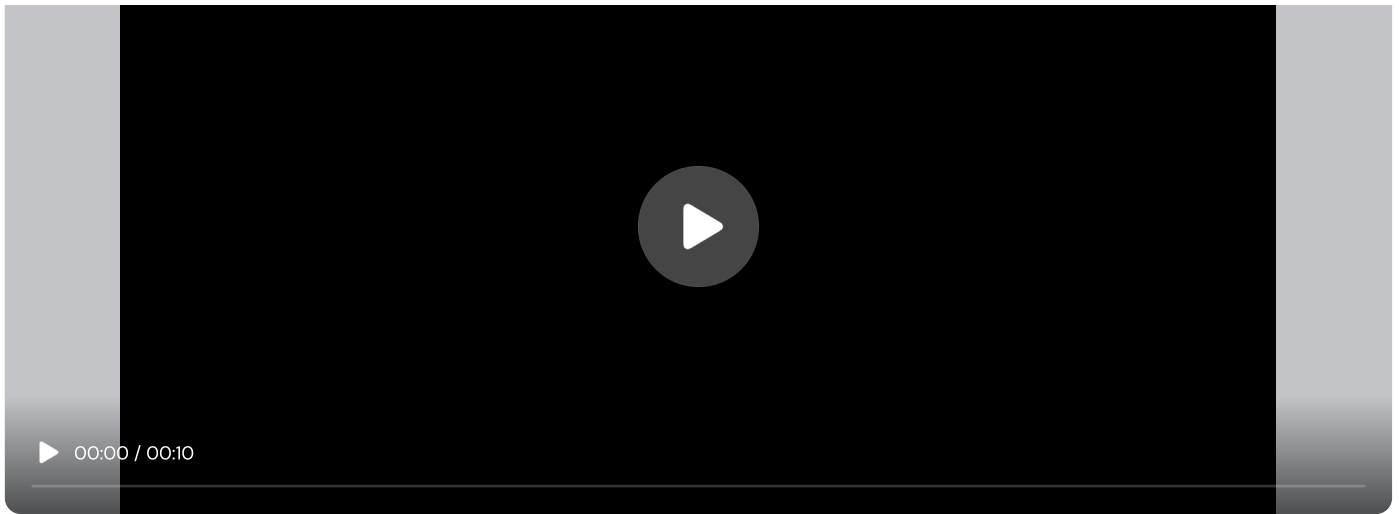
MiniMax-M2.1 also demonstrates steady improvements over M2 in both long-horizon tool use and comprehensive intelligence metrics.

Benchmark	MiniMax-M2.1	MiniMax-M2	Claude Sonnet 4.5	Kimi K2 Thinking	DeepSeek V3.2	GLM 4.6	Claude Opus 4.5	Gemini 3 Pro	GPT-5.2 (thinking)
Toolathlon	43.5	18.7	38.9	17.6	35.2	18.8	43.5	36.4	41.7
BrowseComp	47.4	44.0	19.6	41.5	51.4	45.1	37.0	37.8	66.8
BrowseComp (context management)	62.0	56.9	26.1	60.2	67.6	50.2	57.8	58.2	70.0
AIME25	81.0	78.0	88.0	95.0	92.0	86.0	91.0	96.0	98.0
MMLU-Pro	88.0	82.0	88.0	85.0	86.0	83.0	90.0	90.0	87.0
GPQA-D	81.0	78.0	83.0	84.0	84.0	78.0	87.0	91.0	90.0
HLE w/o tools	22.0	12.5	17.3	22.3	22.2	13.3	28.4	37.2	31.4
LCB	78.0	83.0	71.0	85.0	86.0	70.0	87.0	92.0	89.0
SciCode	39.0	36.0	45.0	42.0	39.0	38.0	50.0	56.0	52.0
IFBench	70.0	72.0	57.0	68.0	61.0	43.0	58.0	70.0	75.0
AA-LCR	62.0	61.0	66.0	66.0	65.0	54.0	74.0	71.0	73.0
V-Bench Telecom	87.0	87.0	78.0	93.0	91.0	71.0	90.0	87.0	85.0

Showcases

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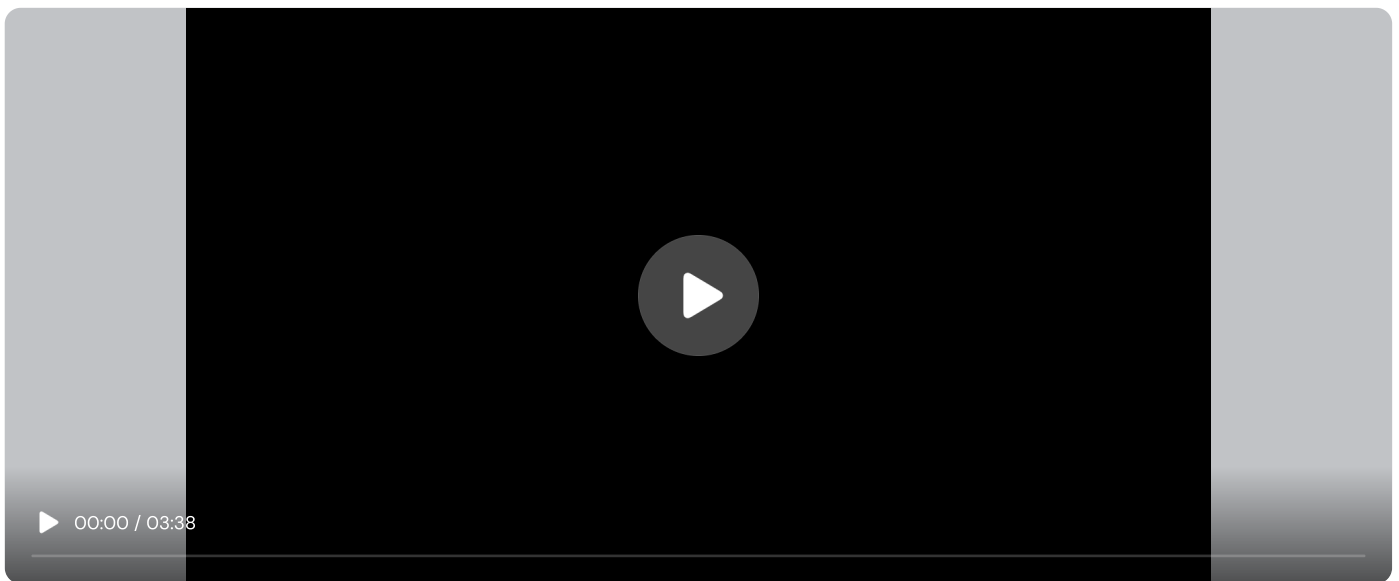
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3D Interactive Animation

MiniMax M2.1 built a "3D Dreamy Christmas Tree" based on React Three Fiber and InstancedMesh, successfully rendering over 7,000 instances. It supports gesture interaction and complex particle animation, demonstrating advanced 3D rendering capabilities. Try it out: <https://yuyl27wq92.space.minimax.io/>

Agentic Tool Use



Tool Use Capability: Excel Market Research

M2.1 demonstrated its tool-use capabilities by autonomously invoking Excel and Yahoo Finance to complete an end-to-end task, ranging from market research data cleaning and analysis to chart generation.

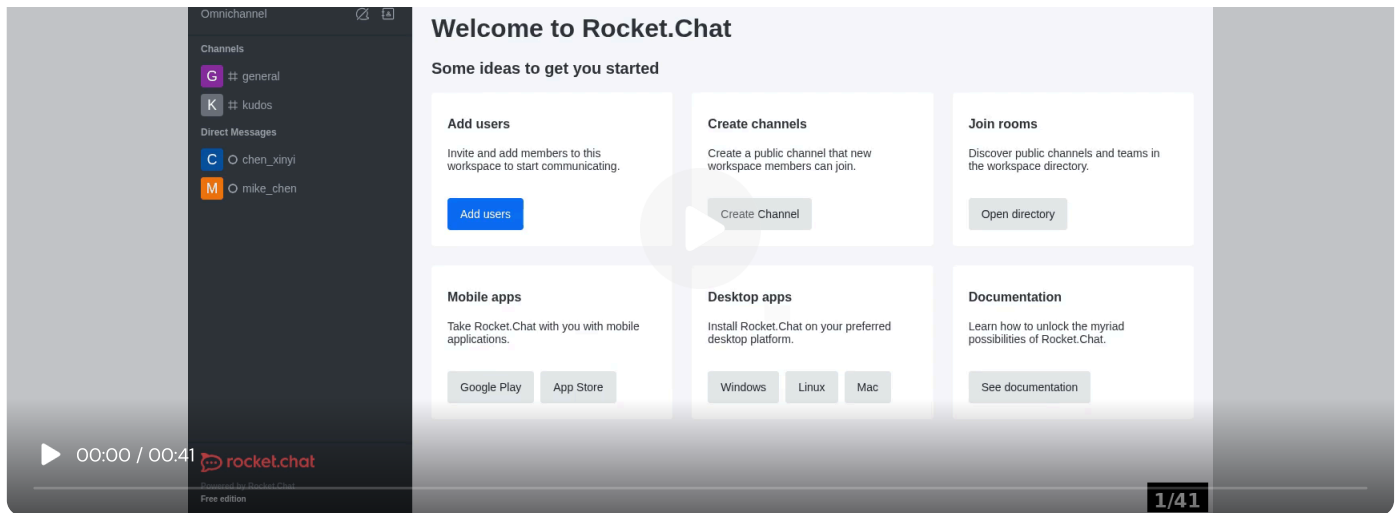
Digital Employee

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Demo 1: Administrative tasks

Task Requirements: Proactively collect employees' equipment requests on communication software, then search for relevant documents on the enterprise's internal server to obtain equipment prices, calculate the total cost and determine whether the department budget is sufficient, and then record equipment changes.

Local Deployment Guide

Download the model from [HuggingFace repository](#)

We recommend using the following inference frameworks (listed alphabetically) to serve the model:

SGLang

We recommend using [SGLang](#) to serve MiniMax-M2.1. Please refer to our [SGLang Deployment Guide](#).

vLLM

We recommend using [vLLM](#) to serve MiniMax-M2.1. Please refer to our [vLLM Deployment Guide](#)

Transformers

We recommend using [Transformers](#) to serve MiniMax-M2.1. Please refer to our [Transformers Deployment Guide](#).

Other Inference Engines

Ktransformers

Inference Parameters

We recommend using the following parameters for best performance: $temperature=1.0, top_p = 0.95, top_k = 40$

Tool Calling Guide

Please refer to our [Tool Calling Guide](#).

How to Use

The MiniMax-M2.1 API is now live on the MiniMax Open Platform: <https://platform.minimax.io/docs/guides/text-generation>

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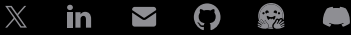
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